WasteLink - Smart Waste Exchange Platform

Submitted by: Mahesh S (1NH25MC064) & Manoj Kumar H M (1NH25MC069)
College: New Horizon College of Engineering, Bengaluru
Academic Year: 2025-2026
Architecture Guide: Model-View-Controller (MVC) Pattern

1. System Architecture & Design Pattern

The WasteLink application is built using a clean separation of concerns under the Model-View-Controller (MVC) pattern:

Model: Java Beans (POJOs) in the *com.wastelink.model* package representing database tables, and Data Access Objects (DAOs) in *com.wastelink.dao* executing parameterized SQL queries.

View: JavaServer Pages (JSP) in *src/main/webapp/* integrating with Bootstrap 5 CSS, Custom animations, and Chart.js dashboards.

Controller: Java Servlets in *com.wastelink.servlet* that intercept client requests, read parameters, run business logic rules, and select the target view redirects.

2. Database Schema Configuration

The SQL schema is stored in **database/wastelink_schema.sql** and consists of 5 related tables:

USERS: Core credentials, contact profiles, user roles (INDUSTRY or RECYCLER), and GPS locations.

WASTE: Contains listings posted by industries (category, quantity in kg, price, location coordinates, and availability status).

RECYCLER: Holds buyer details linked 1-to-1 with USERS (approved category filters, capacities, and compliance certificate numbers).

MATCHES: Holds buyer-seller connections, transaction status logs, profit estimations, and carbon credit metrics.

NOTIFICATIONS: Records system alerts, read/unread states, and logs matching updates.

3. Embedded Core Utility Calculations

Haversine Distance (LocationUtil.java): Computes distances in kilometers relative to the Earth's radius (6371 km) to run spatial matches.

Profit Estimation (ProfitUtil.java): Gross Profit is calculated using the business rule:
*Profit = (Quantity * Price) - (8% handling overhead) - (Rs. 0.5/kg transportation costs)*

CO2 Offsets (ImpactUtil.java): Maps categories to carbon reduction constants (e.g. Plastic = 2.5, Metal = 1.8, Paper = 1.1, E-Waste = 3.2, Chemical = 2.0, Organic = 0.6) and scores sustainability progress.

4. Directory Structures & File Index

Directory Path	File Name	Role / Content Summary
database/	wastelink_schema.sql	Database setup, structures & test data
src/main/webapp/	index.jsp	Landing page with system highlights

	login.jsp / register.jsp	Authentications, role forms & GPS coordinates
	industry-dashboard.jsp	Industry dashboard listing posts & matches
	recycler-dashboard.jsp	Recycler dashboard listing bids & contact logs
	marketplace.jsp	Marketplace search grids & filters
	analytics.jsp	Chart.js statistics & margin calculators
com.wastelink.db	DBConnection.java	MySQL database JDBC connector (Mahesh@123)
com.wastelink.model	User / Waste / Match / Recycler	Encapsulated Java Beans (Getters/Setters)
com.wastelink.dao	UserDAO / WasteDAO / MatchDAO	Parameterized SQL statements & CRUD actions
com.wastelink.servlet	LoginServlet / MatchingServlet	HTTP Session managers, controllers & routing logic
com.wastelink.util	Location / Profit / ImpactUtil	Mathematical formulas for distance, margins, & carbon